

Trainer Debrief — Lab 1

Select and Charter Your Capstone Project

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Select a viable Black Belt project from a candidate portfolio and charter it (A1, A2).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-01-project-selection-and-charter.xlsx

A defensible charter rejects both 'solution in disguise' candidates outright. Scored on strategic fit and benefit, 'Reduce seal weld leak defects' wins — it has the largest COPQ (~SGD 812,000/yr), accessible MES data and a named sponsor. Its problem statement must contain NO cause and NO solution: 'Between Jan and Dec 2025, 2.9% of infusion pump assemblies failed the seal weld burst test across three plants, costing SGD 812,000 in scrap, rework and warranty.' This is the capstone project every later lab builds on.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

You have a weighted selection matrix with at least five scored candidates, a documented rejection rationale, an annualised COPQ figure, and a complete charter whose problem statement contains no cause and no solution.

Where this sits in the capstone

This is the most important lab of the course — everything you build for the next five days attaches to the project you select here. Screen candidate projects the way a Black Belt does: against strategic fit, financial benefit, data availability and risk. Then write the charter that will govern the work. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27

Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30
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If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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